



2.3 Managing Finance

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- * Profit
- * Liquidity
- * Business Failure



Profit calculations

- Profit is the money left over **after all costs have been accounted for**
- There are several different types of profit

Types of profit

Type of profit	What does it show?	How is it calculated?
Gross profit (GP)	The difference between revenue and the costs directly related to production	$GP = \text{revenue} - \text{cost of sales}$
Operating profit (OP)	The difference between the gross profit and the indirect expenses involved in operating the business	$OP = \text{gross profit} - \text{operating expenses}$
Net profit (NP)	The difference between the operating profit and any interest paid and received, as well as any one-off costs	$NP = \text{operating profit} - (\text{net interest} + \text{exceptional costs})$



Worked Example

An e-scooter manufacturer sells its products to retailers for £180 per unit. Variable costs are $\frac{2}{5}$ of the selling price, with monthly fixed costs being £82,000. It sells 2,200 scooters a month.

The business pays £240 interest on a mortgage each month. This year, it purchased the patent for a new type of rechargeable battery for £17,000.

Calculate the business's **net profit** for the year.

[5]

Step 1: Calculate the variable cost per unit

$$= \frac{2}{5} \text{ of } £180$$

$$= £72$$

Step 2: Calculate the gross profit per unit (selling price – variable cost per unit)



Your notes

$$= £180 - £72$$

[1]

$$= £108$$

Step 3: Calculate the gross profit per month (gross profit per unit × units sold)

$$= £108 \times 2,200$$

$$= £237,600$$

Step 4: Calculate the gross profit per year (gross profit per month × 12)

$$= £237,600 \times 12$$

[1]

$$= £2,851,200$$

Step 5: Multiply monthly fixed costs by 12 (months)

$$= £82,000 \times 12 \text{ months}$$

$$= £984,000$$

Step 6: Subtract the annual fixed costs from the annual gross profit

$$= £2,851,200 - £984,000$$

[1]

$$= £1,867,200$$

Step 7: Multiply the monthly interest by 12 (months)

$$= £240 \times 12$$

[1]

$$= £2,880$$

Step 8: Add the one-off purchase to the annual interest

$$= £17,000 + £2,880$$

$$= £19,880$$

Step 9: Subtract the interest and one-off costs from the operating profit



$$\begin{aligned} &= \text{£ } 1,867,200 - \text{£ } 19,880 \\ &\quad [1] \\ &= \text{£ } 1,847,320 \end{aligned}$$



Examiner Tips and Tricks

You may not be asked to complete all of these calculations in one question.

The question may, for example, provide the gross profit and some other information and then ask you to calculate the net profit.

Look at the data carefully to ensure you are doing the correct calculation.

Statement of comprehensive income (profit and loss account)

- The statement of comprehensive income is an end-of-year financial statement that shows all of a business's **income and expenses over the previous 12 months**
- Each type of **profit is calculated** within the statement of comprehensive income
- The previous year's figures are also shown for comparison purposes

Example statement of comprehensive income for Head to Toe Wellbeing Ltd

	2022 (£)	2021 (£)
Revenue	124,653	121,791
Cost of Sales	18,922	22,646
Gross Profit	105,731	99,145
Administrative expenses	39,867	36,448
Operating Profit	65,864	62,697
Interest	2,043	2,952
Exceptional Costs	6,225	0
Profit for the year	57,596	59,745
Taxation	5,470	5,680
Profit for the year after tax	52,126	54,065

GROSS PROFIT =
REVENUE – COST OF SALES

OPERATING PROFIT =
GROSS PROFIT – EXPENSES

PROFIT FOR THE YEAR =
OPERATING PROFIT –
INTEREST & EXCEPTIONAL
COSTS

**PROFIT FOR THE YEAR
AFTER TAX = PROFIT FOR
THE YEAR – TAX**

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Profit margins

- A profit margin is the amount by which the **sales revenue exceeds the costs**
 - Profit margins can be calculated for each type of profit (gross, operating and net profit)
- Profit margins can be **compared to previous years** to better understand business performance
 - **Higher and increasing profit margins are preferable**, as it means that more revenue is being converted to profit

Gross profit margin

- This shows the **proportion of revenue** that is turned into gross profit and is expressed as a percentage
 - It is calculated using the formula

$$\frac{\text{Gross profit}}{\text{Revenue}} \times 100$$



Worked Example

Head to Toe Wellbeing Ltd's revenue in 2022 was £124,653. Its gross profit was £105,731.

Calculate Head to Toe Wellbeing's gross profit margin in 2022.

[2]

Step 1: Substitute the values into the formula

$$\begin{aligned} \text{GPM} &= \frac{\text{Gross profit}}{\text{Revenue}} \times 100 \\ &= \frac{\pounds 105,731}{\pounds 124,653} & [1] \\ &= 0.8482 \end{aligned}$$

Step 2: Multiply the outcome by 100 to find the percentage

$$= 0.8482 \times 100$$

$$= 84.82\%$$



Your notes

Operating profit margin

- The **operating profit margin** shows the proportion of revenue that is turned into operating profit and is expressed as a percentage
- It is calculated using the formula

$$\frac{\text{Operating profit}}{\text{Revenue}} \times 100$$



Worked Example

Head to Toe Wellbeing Ltd's revenue in 2022 was £124,653. Its operating profit was £65,864.

Calculate Head to Toe Wellbeing's operating profit margin in 2022.

[2]

Step 1: Substitute the values into the formula

$$\text{OPM} = \frac{\text{Operating profit}}{\text{Revenue}} \times 100$$

$$= \frac{£ 65,864}{£ 124,653}$$

$$= 0.5284$$

[1]

Step 2: Multiply the outcome by 100 to find the percentage

$$= 0.5284 \times 100$$

$$= 52.84\%$$

[1]

Net profit margin

- The net profit margin (also known as the **profit for the year margin**) shows the proportion of revenue that is turned into net profit before tax and is expressed as a percentage
- It is calculated using the formula

$$\frac{\text{Profit for the year}}{\text{Revenue}} \times 100$$



Your notes



Worked Example

Head to Toe Wellbeing Ltd's revenue in 2022 was £124,653. Its profit for the year was £57,596.

Calculate Head to Toe Wellbeing's profit for the year margin (net profit) in 2022.

[2]

Step 1: Substitute the values into the formula

$$\begin{aligned} \text{NPM} &= \frac{\text{Profit for the year}}{\text{Revenue}} \times 100 \\ &= \frac{£57,596}{£124,653} & [1] \\ &= 0.4621 \end{aligned}$$

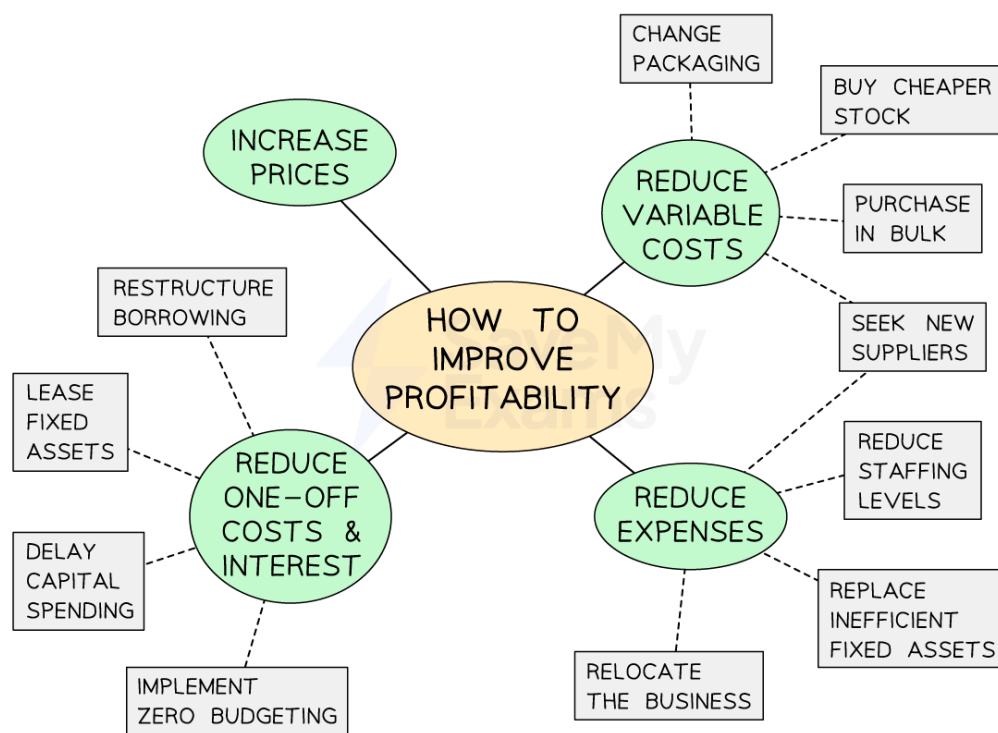
Step 2: Multiply the outcome by 100 to find the percentage

$$\begin{aligned} &= 0.4621 \times 100 \\ & & [1] \\ &= 46.21\% \end{aligned}$$

Ways to improve profitability

- There are several steps a business can take to **improve profitability**

Main ways to improve profitability



Businesses can improve profitability by reducing costs or increasing revenue

Raising prices

- If costs remain the same, raising prices will improve profitability, as the **difference** between the selling price and the cost is now greater
- Raising prices is likely to have an impact on demand, so a business must understand the **price elasticity of demand** for its products
 - Where demand for products is **price elastic**, increasing prices will result in lower revenue — in this case, profitability will be reduced
 - Where demand for products is **price inelastic**, increasing prices will increase revenue — in this case, profitability will rise

Reducing variable costs

- This may involve purchasing cheaper/alternative resources, negotiating with suppliers or **purchasing in bulk**
 - Businesses must ensure that reducing variable costs will not have an **adverse effect on the quality or desirability** of products
 - Buying stock in greater quantities may require investment in **increased storage space**, which will reduce the impact of the cost savings made
 - Businesses may also be able to reduce the **waste** of raw materials and components

Reducing other expenses



- Reducing staffing levels, relocating to cheaper premises or changing utility companies can reduce expenses
 - Reducing staffing levels may affect **staff morale** and negatively affect **productivity**
 - **Relocation costs** can outweigh some of the benefits of moving to a cheaper location
 - Replacing inefficient or outdated equipment may require staff **training**

Reducing one-off costs and interest charges

- Delaying the purchase of fixed assets, entering **leasing** arrangements or restructuring borrowing can reduce costs
 - Delaying purchases of new fixed assets (e.g. machinery or vehicles) may negatively impact **capacity utilisation** as a result of increased breakdowns and maintenance of the old equipment
 - The **leasing** of equipment (e.g. photocopiers) can reduce one-off purchase costs, but the business never owns these **assets**, which weakens the **balance sheet**
 - Restructuring borrowing can result in lower monthly payments but **requires lenders to agree to new terms**, which they may not be willing to do

The distinction between profit and cash

- Profit and cash are different financial terminologies
 - Profit is simply **the difference between revenue generated and business costs**
 - Cash is measured by taking into account the full range of **money flowing in and out of a business**
- A new business may have to **pay cash on purchase for all its supplies** until a good business relationship has built up a level of trust with its suppliers
 - A supplier may then give the business **trade credit** of 30 or 60 days
 - This means that the business can receive its stock now and only pay for it in 30 or 60 days, so the cash outflow is delayed
 - As the business **sells its products**, it receives money generated from the business revenue, and this represents a **cash inflow**
 - **At the end of 60 days, the business will pay its supplier** (cash outflow), but the firm may **still have stock available for sale**
- A profitable business is likely to fail if it **does not have sufficient cash**
 - Cash-poor businesses will struggle to pay their suppliers
 - E.g. lifestyle retailer Joules announced plans to **liquidate** in December 2022 as a result of cash flow difficulties despite making a profit of £2.6 million during the previous year



Statement of financial position (balance sheet)

- The statement of financial position contains the financial information required to draw conclusions about the **liquidity of the business**
 - Liquidity is the ability of a business to **meet its short-term commitments** (e.g. payments to creditors) with its available assets
 - A business that cannot pay its bills will usually **fail very quickly**, even if it is profitable
 - Managing liquidity is a key way to **manage risk** in a business — and it helps a business to prepare for the unexpected
- The statement of financial position shows the **financial structure** of a business at a specific point in time
 - It identifies a business's **assets** and **liabilities** and specifies the **capital** (money) used to fund the business
 - The statement of financial position is also known as the **balance sheet**

Example of a statement of financial position for Packer Sports Ltd



Your notes

	£	£	NON-CURRENT ASSETS ARE ITEMS THAT ARE OWNED BY A BUSINESS FOR THE LONG-TERM. EXAMPLES INCLUDE MACHINERY AND BUILDINGS
Non-current Assets			
Vehicles	42,000		
Machinery	18,500	60,500	
Current Assets			CURRENT ASSETS ARE ITEMS THAT ARE CONVERTED TO CASH QUICKLY – USUALLY WITHIN 12 MONTHS. CURRENT ASSETS ARE COMPRISED OF CASH, TRADE RECEIVABLES AND INVENTORY
Inventory	8,250		
Trade Receivable	2,630		
Bank	4,080		
Cash	585	15,545	
Current Liabilities			CURRENT LIABILITIES IS MONEY A BUSINESS OWES AND IS DUE TO BE SETTLED SOON – USUALLY WITHIN 12 MONTHS. CURRENT LIABILITIES INCLUDE TRADE PAYABLES AND SHORT-TERM BORROWING SUCH AS A BANK OVERDRAFT
Trade Payables	3,960		
Bank Overdraft	1,100	5,060	
Net Current Assets		10,485	
Non-current Liabilities			NON-CURRENT LIABILITIES IS MONEY A BUSINESS OWES AND THAT DOES NOT NEED TO BE PAID BACK FOR AT LEAST 12 MONTHS. NON-CURRENT LIABILITIES INCLUDE BANK LOANS AND MORTGAGES
Mortgage		20,000	
Net Assets		50,985	NET ASSETS = ASSETS – LIABILITIES
Financed by			THE STATEMENT OF FINANCIAL POSITION SHOWS HOW THE NET ASSETS OF A BUSINESS ARE FUNDED. IN THIS CASE THE BUSINESS IS FUNDED WITH A COMBINATION OF OWNERS CAPITAL AND PROFIT GENERATED BY THE BUSINESS. THE TOTAL FUNDING IS KNOWN AS THE CAPITAL EMPLOYED. NET ASSETS = CAPITAL EMPLOYED
Capital		16,595	
Profit for the year		36,990	
Less Drawings		2,600	
Capital Employed		50,985	

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Ways to measure liquidity

- The liquidity of a business can be measured using two ratios, the **current ratio** and the **acid test ratio**

The current ratio



Your notes

- The current ratio is a **quick way** to measure liquidity, and the outcome is **expressed as a ratio**
- All forms of current assets are considered in this ratio
- The current ratio is an effective liquidity measure for businesses that **hold little stock**
- The result indicates how many **£s of current assets** a business has available to **cover each £1 of its short-term debt**
- It is calculated using the formula

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}} = ? : 1$$



Worked Example

Packer Sports Ltd has current assets of £15,545, current liabilities of £5,060 and an inventory figure of £8,250.

Calculate Packer Sports Ltd's current ratio.

[2]

Step 1: Substitute the values into the equation

$$\begin{aligned} &= \frac{£ 15,545}{£ 5,060} \quad [2] \\ &= 3.07 : 1 \end{aligned}$$

In this example, Packer Sports Ltd has **£3.07** of current assets to cover each **£1** of short-term debt.

The acid test ratio

- This is a precise way to measure liquidity and is **expressed as a ratio**
- The acid test ratio is also known as the **liquid capital ratio**
- The **least liquid** form of current assets (inventory/stock) is deducted, so the acid test ratio provides a more realistic measure of the business's ability to **meet short-term debts quickly**
- It is a particularly important measure of liquidity for businesses that **hold a large amount of stock**
- It is calculated using the formula

$$\text{Acid test} = \frac{\text{Current assets} - \text{Inventory}}{\text{Current liabilities}} = ? : 1$$



Your notes

Worked Example

Packer Sports Ltd has current assets of £15,545, current liabilities of £5,060 and an inventory figure of £8,250.

Calculate Packer Sports Ltd's acid test ratio.

[2]

Step 1: Subtract the inventory figure from the current assets

$$\begin{aligned} &= £15,545 - £8,250 \\ &= £7,295 \end{aligned} \quad [1]$$

Step 2: Substitute the values into the equation

$$\begin{aligned} &= \frac{£7,295}{£5,060} \quad [1] \\ &= 1.44 : 1 \end{aligned}$$

In this example, Packer Sports Ltd has £1.44 of its most liquid current assets to cover each £1 of short-term debt.

Ways to improve liquidity

- The best way to improve liquidity is to **manage the business better**
 - Use **cash flow forecasts** to identify potential cash flow issues before they arise — and take appropriate action
 - **Budget** effectively and consider adopting **zero budgeting** to carefully control spending
 - Set clear **financial objectives** and look for ways to **reduce costs** and **increase income** wherever possible

How to improve liquidity

Method	Explanation
Reduce the credit period offered to customers	▪ Collecting money owed from customers more quickly will increase the level of current assets in the business ▪ Customers may move to competing businesses that offer better credit terms



Your notes

Ask suppliers for an extended repayment period, e.g. an extension from 60 to 90 days	▪ Current liabilities will not be reduced ▪ The business can use the cash it would have paid to suppliers for other purposes ▪ Suppliers may be unwilling to extend credit terms
Make use of overdraft facilities or short-term loans	▪ Current liabilities will increase ▪ The business can spend more money than it has in its bank account ▪ Banks may be reluctant to lend to businesses with cash-flow problems
Sell off excess stock	▪ Less liquid current assets will be reduced and converted into more liquid forms of current assets (e.g. cash) ▪ Storage and security costs may also be reduced ▪ Stock may need to be sold at a low price to attract sales
Sell assets and lease fixed assets instead (e.g. sale and leaseback)	▪ Both current assets and current liabilities will increase ▪ The business will continue to have the use of the assets but must make regular payments to the leasing company
Introduce new capital and reduce drawings from the business	▪ New capital may be introduced by the owner or from additional investors ▪ This may result in the dilution of control of the business

Managing working capital

- Working capital is the money that a business has to fund its day-to-day activities
- It is often described as **net current assets** on the statement of financial position
- Working capital is calculated using the formula

Working capital = Current assets – Current liabilities



Worked Example

Garrick Components Ltd is a heating components business based in Tamworth. It has been struggling to control its inventory level. Its customers are the country's leading gas boiler manufacturers. These companies require Garrick Components Ltd to supply products "just in time", and as a result, Garrick must hold large amounts of varied stock to ensure that its customers' needs can be met. Garrick Components offers its customers 90-day credit terms.

Financial information for Garrick Components Ltd

	2022 (£m)	2021 (£m)
Inventories	8.1	7.2
Trade receivables	2.2	3.1
Cash	0.9	1.2
Short-term loan	6.4	4.4
Trade payables	5.1	5.9

Calculate the difference between Garrick Components Ltd's working capital in 2021 and 2022.

[4]

Step 1: Identify and calculate the current assets and current liabilities for 2022 and 2021

$$\begin{aligned}\text{Current assets } 2022 &= \text{£ } 8.1\text{m} + \text{£ } 2.2\text{m} + \text{£ } 0.9\text{m} = \text{£ } 11.2\text{m} \\ 2021 &= \text{£ } 7.2\text{m} + \text{£ } 3.1\text{m} + \text{£ } 1.2\text{m} = \text{£ } 11.5\text{m}\end{aligned}\quad [1]$$

$$\begin{aligned}\text{Current liabilities } 2022 &= \text{£ } 6.4\text{m} + \text{£ } 5.1\text{m} = \text{£ } 11.5\text{m} \\ 2021 &= \text{£ } 4.4\text{m} + \text{£ } 5.9\text{m} = \text{£ } 10.3\text{m}\end{aligned}\quad [1]$$

Step 2: Subtract the current liabilities from the current assets for 2022 and 2021

$$\begin{aligned}2022 &= \text{£ } 11.2\text{m} - \text{£ } 11.5\text{m} = (\text{£ } 0.3\text{m}) \\ 2021 &= \text{£ } 11.5\text{m} - \text{£ } 10.3\text{m} = \text{£ } 1.2\text{m}\end{aligned}\quad [1]$$

Step 3: Calculate the difference between 2022 and 2021

$$\begin{aligned}&= \text{£ } 1.2\text{m} - (\text{£ } 0.3\text{m}) \\ &= \text{£ } 1.5\text{m}\end{aligned}\quad [1]$$



Your notes

Managing working capital



- Working capital is described as the **lifeblood of a business** because a lack of it often leads to **business failure**, as a business **cannot meet its immediate financial obligations**
 - **Cash is the most liquid** of a business's current assets and can be used to settle **debts** immediately
- The effective management of working capital involves careful cash management
 - Debtors and inventory (e.g. stock) are **less liquid**
 - Businesses that are struggling with a lack of working capital may look to **convert these current assets into cash as quickly as possible** (e.g. by selling the stock at lower prices or by more purposefully chasing payment from customers)
 - Requesting an extension of **payment terms** from suppliers can increase working capital in the short term, as cash remains in the business for longer
 - Making use of **short-term borrowing options** such as **overdrafts** can improve a business's working capital situation, as it **can access more cash** than it has in its current account
- A business can have **too much working capital**
 - If a business is **holding large amounts of cash**, it is likely to be **missing out** on the benefits of investing it in fixed assets or investments
 - This may represent a significant **opportunity cost**, especially when interest rates are high
 - If a business is **holding large amounts of inventory**, it may incur **extra storage costs** (e.g. security and handling costs) and could use the cash "tied up" in this stock for other purposes



Examiner Tips and Tricks

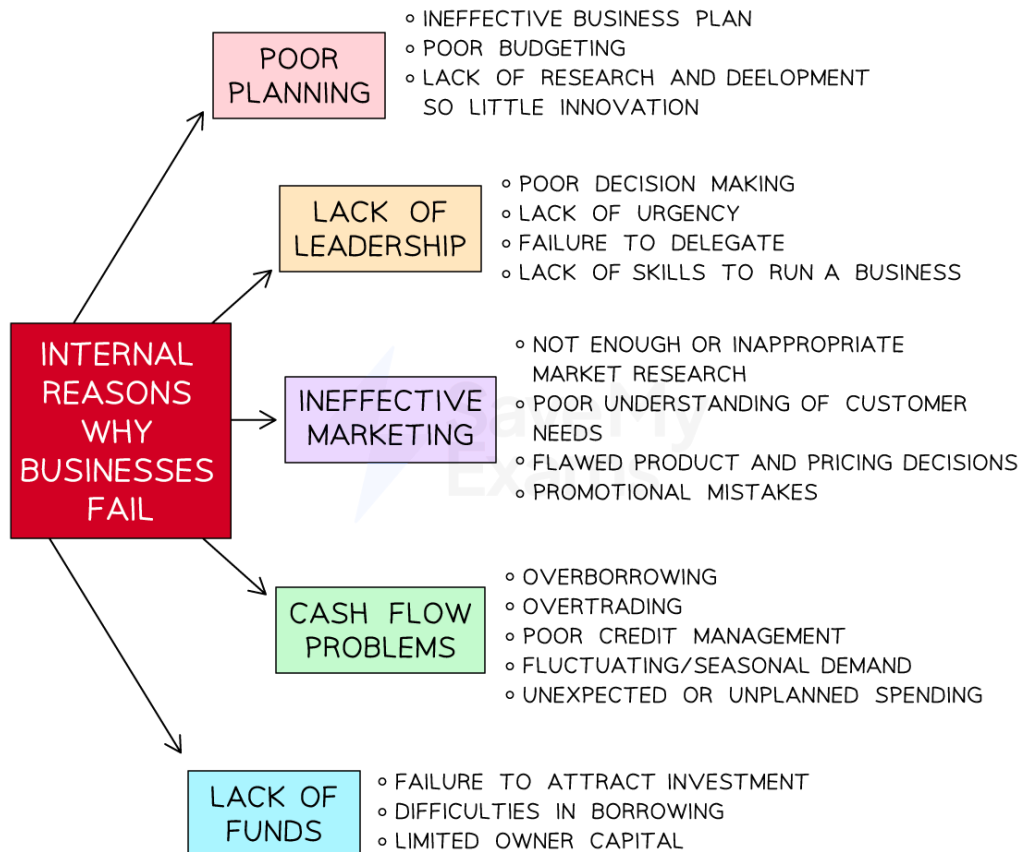
A common exam error is the confusion between working capital and cash. Whilst working capital includes cash, it also includes less liquid current assets (e.g. trade receivables and inventory).

These less liquid assets cannot be used to pay bills, so whilst a business may have a positive working capital figure, it may still fail because it cannot meet its immediate financial commitments.



Internal and external causes of business failure

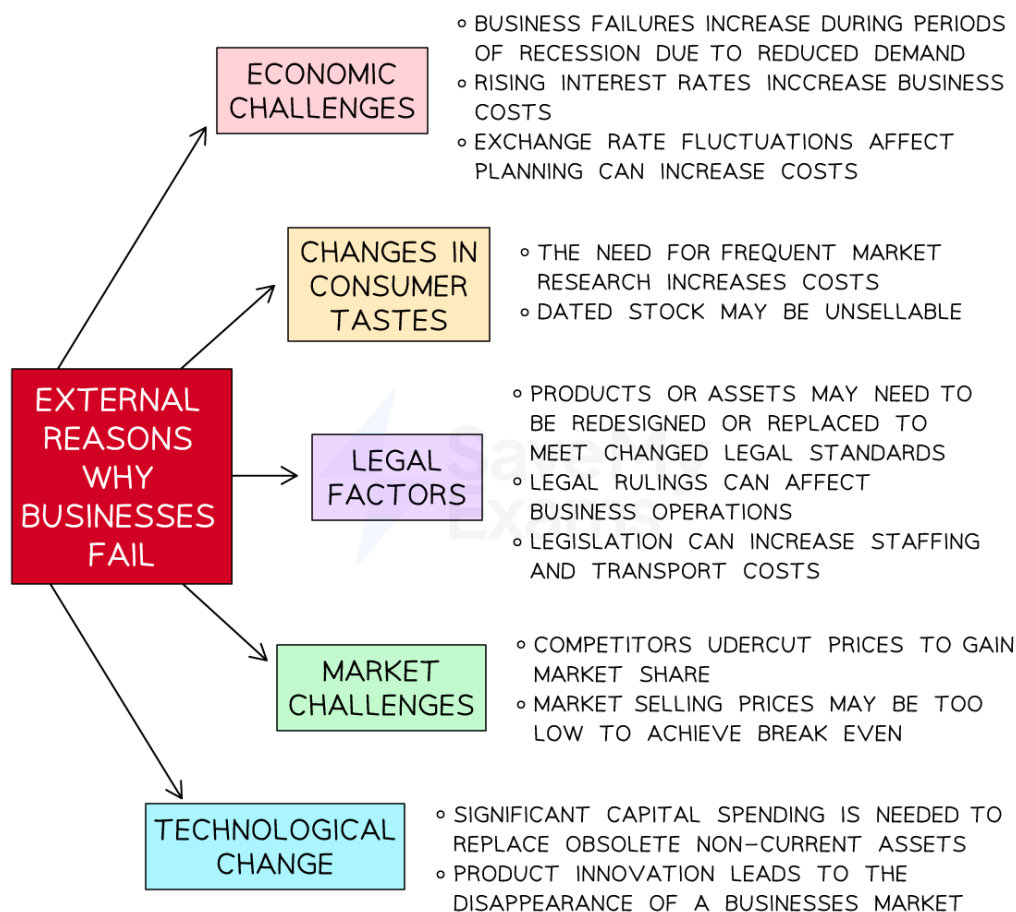
Internal causes of business failure



The main internal reasons for business failure include ineffective marketing, lack of leadership and lack of funds

- **Ineffective management** is a key cause of business failure
- Leaders may lack the **experience or skills** to run a business effectively, **especially during periods of crisis**
- Financial factors (e.g. a **lack of cash flow** or **working capital**) can cause swift business failure, as it becomes difficult to operate on a **day-to-day** basis

External causes of business failure



The main external reasons for business failure include changes in consumer taste, economic challenges and market challenges

- Businesses also fail as a result of factors **beyond their direct control**
- A **change in legislation** can mean that products or processes may require significant redesign or withdrawal
 - E.g. the UK government announced that diesel vehicles are banned from 2030. Car manufacturers are switching to the production of greener petrol and electric vehicles, and some have moved production processes outside of the UK to reduce costs
 - Several prominent vehicle manufacturers **will likely fail** in the future as a result of these changes
- **Economic challenges** (e.g. **rising interest rates** or a **recession**) are a key cause of business failure
 - This is especially true for businesses that rely heavily on borrowing or that sell items that require customers to borrow money to finance their purchase
- The **entry of new competitors** into a market can cause significant problems for incumbent businesses, who may have to **slash prices or spend heavily on promotional activity** to maintain their market share

- If a business has **existing weaknesses**, the entry of a new competitor can hasten its failure



Your notes